(ii) $f(y) = 6y^2 - 7y + 2$ $\quad g(y) = 7y + y^3$

(iii) $h(z) = z^5 - 6z^4 + z$ $\quad f(z) = 6z^2 + 10z - 7$

7. What should be added to $2x^3 + 6x^2 - 5x + 8$ to get $3x^3 - 2x^2 + 6x + 15$?

8. What must be subtracted from $2x^4 + 4x^2 - 3x + 7$ to get $3x^3 - x^2 + 2x + 1$?

9. Multiply the following polynomials and find the degree of the resultant polynomial:

(i) $p(x) = x^2 - 9$ $\quad q(x) = 6x^2 + 7x - 2$

(ii) $f(x) = 7x + 2$ $\quad g(x) = 15x - 9$

(iii) $h(x) = 6x^2 - 7x + 1$ $\quad f(x) = 5x - 7$

10. The cost of a chocolate is Rs. $(x + y)$ and Amir bought $(x + y)$ chocolates. Find the total amount paid by him in terms of x and y. If x =10, y =5 find the amount paid by him.

11. The length of a rectangle is $(3x+2)$ units and it's breadth is $(3x-2)$ units. Find its area in terms of x. What will be the area if $x = 20$ units.

12. $p(x)$is a polynomial of degree 1 and $q(x)$ is a polynomial of degree 2. What kind of the polynomial $p(x) \times q(x)$ is ?

3.2.2 Value and Zeros of a Polynomial

Consider the two graphs given below. The first is linear, the second is quadratic. The first intersects the X axis at one point ($x = -3$) and the second at two points ($x = -1$ and $x = 2$). They both intersect the Y axis only at one point. In general, every polynomial has a graph and the graph is shown as a picture (since we all like pictures more than formulas, don't we?). But also, the graph contains a lot of useful information like whether it is a straight line, what is the shape of the curve, how many places it cuts the x-axis, etc.

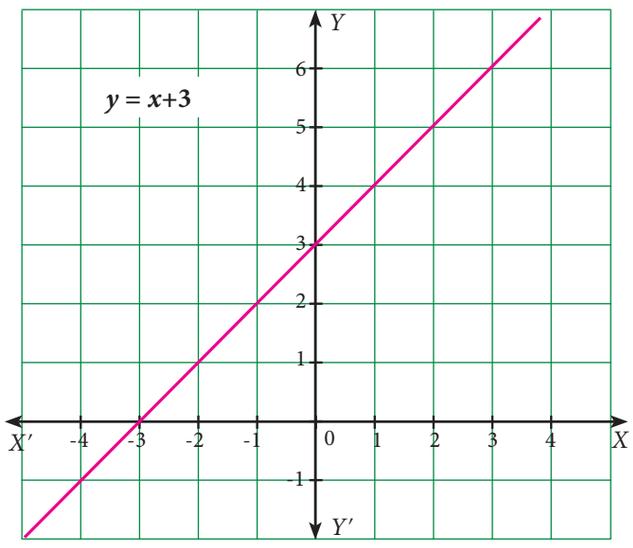


Fig. 3.7

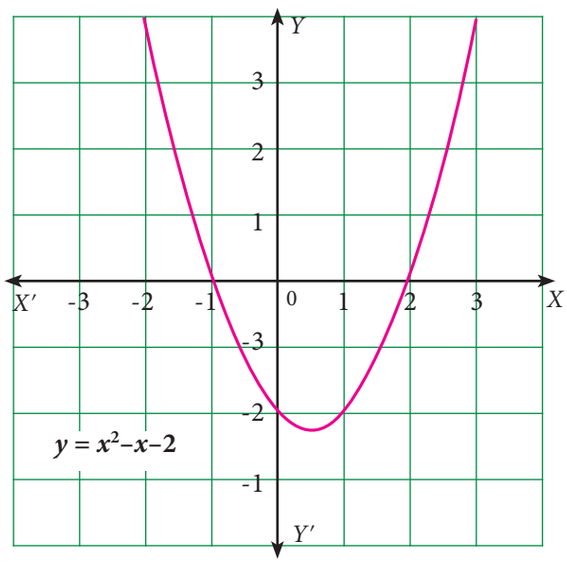


Fig. 3.8

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